

Bonus Homework 1

Discrete Mathematics by Hongfei Fu, 2023.9.28

Name:_____ Student ID:_____ Class: _____

1. (Bonus Homework, **0.2pt**) Prove that $\{\neg, \oplus\}$ is not functional complete.

Hint: First prove from the commutativity, associativity of $\oplus$ and the law $\neg(p \oplus q) \equiv (\neg p) \oplus q$ that any propositional formula that uses only $\neg$ and $\oplus$ can be equivalently transformed into a group- $\oplus$ of propositional variables p and their negations $\neg p$. Then eliminate duplicate propositional variables by using $p \oplus (\neg p) \equiv \mathbf{T}$ and $p \oplus p \equiv \mathbf{F}$. Finally, it should be obvious that the truth of such a propositional formula depends only on the number of propositional variables taking the truth value $\mathbf{T}$, so that conjunction and disjunction cannot be expressed by such propositional formula.

Answer Area: